

Aaron Lilla

Full-Stack Software Engineer

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Authorized to work in the U.S. (no sponsorship required) | Native English speaker

SUMMARY

Full-stack software engineer with 13 years in web development and 8+ years in modern JavaScript and TypeScript. Lead front-end engineer on TableCaptain - a React and Electron platform in 30+ poker rooms across the U.S. and Asia - driving 450+ concurrent in-venue displays over one real-time WebSocket fabric with zero major downtime in six years. Founder of Afterimage Studio: self-published Spire of Ash, a commercial TypeScript/Electron game, on Steam, and building Holoscene on a TypeScript design studio plus a deterministic C# engine with an AI-agent tooling pipeline.

SKILLS

Languages: TypeScript, JavaScript (ES2024), Node.js, C# (.NET), HTML5, CSS3, SQL

Frontend: React, Redux, Next.js, Tailwind CSS, Vite, Three.js, Design Systems, Responsive UI, Accessibility (WCAG)

Backend & Real-time: Node.js, Express, REST APIs, WebSockets, Supabase, PostgreSQL, SQLite, Electron (IPC, preload, auto-update), Auth & Sessions (OAuth)

AI & Tooling: Claude API, MCP (Model Context Protocol), Multi-agent Pipelines, LLM Streaming, Prompt Engineering, AI-assisted Development

Architecture & Practice: System Design, Scalable Architecture, Deterministic Simulation, Code Review, Mentoring, Agile / Scrum

Testing & DevOps: Vitest, Jest, Playwright, CI/CD, Visual Regression, Performance Profiling, Git, AWS, Azure

EXPERIENCE

Founder and Lead Engineer

2025 - Present

Afterimage Studio

Remote

Holoscene (in development) - a buildcraft ARPG built end-to-end across a TypeScript/React design studio and a deterministic C#/.NET engine on Godot, with an AI-agent pipeline that lets Claude prototype and validate changes in-process.

- Built a TypeScript + React (React 19, Vite) design studio as the single source of truth - a deterministic combat simulator with an ability-priority (APL) evaluator, aura and synergy systems, and content authoring for 100 characters, 500 skills, and 2,000+ items.
- Engineered a deterministic, tick-based simulator (seeded RNG, typed event-stream architecture) and a Monte Carlo parity harness that holds the C#/.NET engine within 0.5% of the TypeScript reference across 1,000 seeded runs.
- Wrote 66 automated balance and parity audit scripts (Node.js) that enforce design contracts in CI - per-character DPS/EHP bands, cross-mode parity, loot and affix coverage - plus a 30-suite Vitest pack over the simulator.
- Designed a multi-agent build pipeline of specialized Claude Code agents (combat runtime, balance validator, data exporter, mode implementor, cross-mode validator) coordinating against a frozen invariant contract, with an MCP-driven dashboard monitoring agent and engine state.

Technologies: TypeScript, React, Node.js, Vite, C#, .NET, Godot, MCP, Claude API, Vitest

Founder and Lead Engineer

2022 - 2024

Afterimage Studio

Remote

Spire of Ash (shipped on Steam) - a commercial roguelite ARPG I designed, built, and self-published end-to-end as a TypeScript and Electron desktop app (~74,000 lines of TypeScript, 600+ tests).

- Shipped a full live-service product to Steam and maintained it across 20+ releases (Steamworks integration, electron-builder packaging, staged auto-update, Sentry crash reporting, and Chinese localization).
- Built the 3D game client with Three.js / React Three Fiber and a Pixi.js Path-of-Exile-style passive tree, on a React + Redux Toolkit / Zustand UI with a spatial-hash and pathfinding engine.
- Built a Supabase cloud backend (auth, cloud saves with conflict resolution, IndexedDB fallback) and real-time WebSocket leaderboards ranked per mode and key level.
- Engineered the procedural systems - a 442-affix item generator, a procedural passive-tree generator, and 988 collectible cards - and gated every release with a headless combat worker and Monte Carlo balance audits.

Technologies: TypeScript, React, Electron, Three.js, Pixi.js, Redux, Supabase, Steamworks, Vite, Vitest, Playwright

Frontend Engineer

2019 - 2025

PokerAtlas

Las Vegas, NV (Hybrid)

Led front-end development of TableCaptain, a React and Electron poker room management platform adopted in 30+ rooms across the U.S. and Asia. Video walkthroughs of the Floor Editor, Table Manager, Waitlist, and in-venue TV display at aaronlilla.github.io.

- Architected the real-time WebSocket sync layer - 450+ concurrent displays (15+ per venue across 30+ venues) with sub-second latency - and scaled TableCaptain V2 to the #1 poker room management system by adoption in 2024 (per company assessment).
- Engineered core features end-to-end (floor editor, table manager, waitlist, employee manager, and a real-time in-venue display HUD with optimistic UI and server-confirmed reconciliation); integrated a .NET backend over WebSockets.
- Automated the Electron auto-update and CI/CD pipeline with staged rollout and rollback - bi-weekly releases, sub-10-minute hotfixes, and zero major downtime incidents across six years of production.
- Led the TypeScript and Redux migration, mentored engineers, and introduced code-review practices; translated Figma mockups into reusable React design-system components.

Technologies: React, Redux, Electron, WebSockets, .NET (integration), Webpack, CI/CD, Figma

Frontend Developer

2017 - 2019

USCRE Online

Las Vegas, NV (Hybrid)

Built a single-page React app delivering real-time commercial real estate data for 5,000+ properties.

- Cut quote retrieval time by 50% via Redux-driven caching, request deduplication, and search-path query optimization.
- Shipped a 20-component reusable UI library (~40% fewer front-end bugs in six months) and drove load time under 1.2s into green Core Web Vitals.

Technologies: React, Redux, JavaScript, HTML5, CSS3

PROJECTS

FiveStack

React | Node/Express | Supabase (PostgreSQL) | OAuth

Full-stack Mythic+ team-building platform for World of Warcraft. React 18 front end (React Query, Tailwind) on a Node/Express API with Supabase PostgreSQL (row-level security, real-time subscriptions) and Discord + Battle.net OAuth. AI matchmaking scores players across five compatibility factors (role, rating, experience, schedule, item level) from live Raider.io data. github.com/aaronlilla/FiveStack

Clippa

Electron | React | MCP | ffmpeg | SQLite

Local-first Electron app that finds highlight moments in long-form video for review and clip export. Multi-stage media pipeline (ingest, audio, scene detection, event classification) on a SQLite job queue, with a dual-transport MCP server (HTTP + stdio, bearer-token auth) that lets external agents drive the whole workflow.

EDUCATION

College of Southern Nevada - Organic Chemistry coursework, Las Vegas, NV